

UBON RATCHATHANI AGROMET, Thailand

WMO: 484080

Lat: 15.233N

Lon: 105.033E

Elev: 130

StdP: 99.77

Time Zone: 7.00 (E07)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	15.0	16.3	10.5	8.0	18.8	11.7	8.7	19.3	6.5	22.9	5.8	22.7	1.0	N/A	0.306

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	10.3	37.5	25.9	36.3	25.8	35.3	25.7	27.7	32.9	27.4	32.5	27.1	32.1	0.7	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.5	22.4	29.9	26.2	21.9	29.6	25.9	21.6	29.3	89.2	33.0	87.8	32.6	86.6	32.3	30.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 5.1	4.0	3.2	DB	12.9	39.0	1.2	1.5	12.0	40.0	11.3	40.9	10.6	41.8	9.7	42.9	(4)
(5)			WB	11.4	28.7	1.2	1.0	10.5	29.3	9.7	29.9	9.1	30.4	8.2	31.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	27.6	24.5	26.4	28.7	30.0	29.5	29.0	28.2	28.1	27.9	27.7	26.9	24.8	(6)
(7)		DBStd	2.51	2.32	2.66	2.42	2.02	1.65	1.31	1.42	1.12	1.31	1.19	1.81	2.39	(7)
(8)		HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)
(10)		CDD10.0	6437	449	459	581	601	605	570	563	561	536	548	506	458	(10)
(11)		CDD18.3	3396	191	226	322	351	347	320	305	303	286	289	256	199	(11)
(12)		CDH23.3	36014	1711	2407	3769	4379	4069	3649	3211	3152	2862	2770	2346	1689	(12)
(13)		CDH26.7	15604	699	1189	1955	2331	1908	1531	1183	1099	982	1053	985	689	(13)
(14)	Wind	WSAvg	0.7	1.0	0.8	0.7	0.6	0.5	0.6	0.6	0.5	0.4	0.6	1.1	1.2	(14)
(15)	Precipitation	PrecAvg	1930	6	13	38	78	229	269	417	401	347	112	21	7	(15)
(16)		PrecMax	2328	36	48	105	162	400	473	698	712	736	298	59	86	(16)
(17)		PrecMin	1403	0	0	6	19	114	131	171	232	180	9	0	0	(17)
(18)		PrecStd	275	10	13	27	45	74	92	137	114	126	64	18	16	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.1	36.5	38.3	39.9	37.7	36.0	35.2	33.9	34.1	34.0	34.7	34.6	(19)
(20)			MCWB	23.6	24.4	25.4	25.8	26.4	26.9	26.5	27.2	27.4	26.2	25.8	24.6	(20)
(21)		2%	DB	33.1	35.3	36.8	38.0	36.3	34.7	33.7	33.0	33.0	33.1	33.7	33.3	(21)
(22)			MCWB	23.3	24.2	25.3	26.2	26.6	26.9	26.6	26.9	26.9	25.8	25.5	24.2	(22)
(23)		5%	DB	31.9	34.2	35.7	36.5	35.1	33.6	32.6	32.2	32.2	32.5	32.8	32.2	(23)
(24)			MCWB	22.8	23.9	25.0	26.1	26.6	26.7	26.5	26.7	26.7	25.7	25.1	23.7	(24)
(25)		10%	DB	30.7	32.9	34.5	35.0	33.7	32.5	31.6	31.2	31.2	31.6	31.7	30.7	(25)
(26)			MCWB	22.2	23.4	24.8	25.9	26.5	26.5	26.3	26.4	26.5	25.5	24.6	23.0	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	25.5	26.2	26.7	27.9	27.9	27.9	27.7	27.9	27.7	27.2	27.1	25.7	(27)
(28)			MCD B	31.5	33.5	34.3	35.7	33.9	32.7	32.3	32.5	32.1	31.6	32.5	32.1	(28)
(29)		2%	WB	24.6	25.6	26.2	27.2	27.4	27.4	27.1	27.4	27.4	26.7	26.4	25.0	(29)
(30)			MCD B	30.5	32.7	34.0	34.5	33.3	32.3	31.8	31.6	31.6	31.1	31.8	31.6	(30)
(31)		5%	WB	23.9	25.1	25.8	26.8	27.1	27.1	26.7	27.1	27.1	26.4	25.9	24.3	(31)
(32)			MCD B	29.4	32.0	33.5	33.7	32.8	32.1	31.1	31.0	31.1	30.6	31.1	30.7	(32)
(33)		10%	WB	23.2	24.4	25.4	26.4	26.7	26.8	26.5	26.7	26.7	26.0	25.3	23.6	(33)
(34)			MCD B	29.0	30.9	32.8	33.0	32.1	31.5	30.6	30.3	30.3	30.1	30.1	29.6	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	12.7	13.2	11.6	10.3	8.5	7.3	6.6	6.3	6.5	8.0	9.6	11.5	(35)
(36)			MCDBR	13.4	13.7	12.6	11.7	10.3	8.7	8.0	7.6	7.9	9.3	10.6	12.5	(36)
(37)			MCWBR	4.8	4.4	3.8	3.0	2.7	2.6	2.7	2.8	2.9	3.4	3.7	4.5	(37)
(38)		5% WB	MCDBR	11.7	11.7	11.1	10.3	9.0	7.8	7.2	7.0	7.5	8.2	9.5	11.6	(38)
(39)			MCWBR	4.6	3.9	3.6	3.1	2.9	2.8	2.9	2.9	2.9	3.2	3.6	4.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.458	0.581	0.662	0.610	0.498	0.465	0.461	0.461	0.492	0.500	0.447	0.438	(40)	
(41)		taud	2.168	1.832	1.691	1.854	2.195	2.303	2.313	2.305	2.208	2.159	2.294	2.267	(41)	
(42)		Ebn at Noon	831	743	697	735	810	829	834	843	820	803	837	840	(42)	
(43)		Edh at Noon	145	211	249	212	148	132	131	133	146	150	127	128	(43)	
(44)	All-Sky Solar Radiation	RadAvg	5.15	5.43	5.33	5.56	5.46	5.03	4.72	4.53	4.56	5.01	4.99	4.89	(44)	
(45)		RadStd	0.25	0.24	0.18	0.25	0.27	0.28	0.42	0.32	0.43	0.37	0.21	0.19	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (11 neighbors)	+0.29	N/A	N/A	N/A	+0.22	+0.20	N/A	N/A	+103	+102

Nomenclature: See separate page